Thijmen **Nijdam**

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Professional Summary

Robot Learning Research Engineer at the University of Amsterdam with full ownership of a real-to-sim-to-real pipeline for robot manipulation, in collaboration with Toyota Motor Europe and the Toyota Research Institute. Experienced in imitation learning and VLA policies on real hardware and in training on multi-GPU HPC clusters, using Python, PyTorch, ROS 2, and LeRobot. MSc thesis on world models for robot planning with model predictive control. Co-authored six publications, including papers at ICML and SIGIR.

Experience

Robot Learning Research Engineer, *University of Amsterdam* **May 2026 - Present**

- Training and deploying VLA and imitation learning policies on real hardware, achieving successful rollouts from a single demonstration
- Owning the full robotics stack for a Franka Research 3 arm, from low-level control and data processing to policy deployment
- Collecting precise teleoperated demonstrations and conducting real-world evaluations to identify and resolve policy failure modes
- Reducing the sim-to-real gap by aligning a digital twin with the real scene, so that RL-fine-tuned policies transfer to the real robot

Research Intern, *University of Amsterdam* **Jan. 2024 - Feb. 2026**

- MSc thesis: designed and trained a local world model that manipulates unseen drawers and doors via MPC (RSS Workshop 2026)
- Built segmentation and tracking pipelines to benchmark physical reasoning of video generation models (ICML 2026)
- Benchmarked 40+ information retrieval models, including LLM re-rankers and fine-tuned models, on negation (SIGIR 2025)
- Developed a conformal prediction method for uncertainty quantification in time series forecasting (COPA 2024)

AI Developer, *LeerLevels* **Apr. 2022 - Mar. 2026**

- Built a RAG-based assistant that enables teachers to assemble lessons and edit educational content
- Deployed an LLM-powered grading feature with monitoring via Langfuse, delivering tailored feedback to hundreds of students

AI Engineer, *RoboCup Student Robotics Team* **Feb. 2023 - Jan. 2024**

- Implemented a real-time ball and robot detection pipeline (Single Shot Detector) on NAO humanoid robots
- Developed a whistle detection model for referee signal recognition

Education

MSc Artificial Intelligence **Sep. 2023 - Feb. 2026**

University of Amsterdam

- Grade: 8.3
- Relevant courses: Computer Vision, Reinforcement Learning, Foundation Models, Machine Learning I & II, Deep Learning I & II

BSc Artificial Intelligence **Sep. 2020 - Jun. 2023**

University of Amsterdam

- Grade: 8.1, graduated with honours
- Minor in Computer Science; exchange at the University of Hong Kong

Selected Publications

Local World Models for Articulated Object Manipulation **RSS Workshop**

Robot World Models Workshop @ Robotics: Science and Systems | *First author* 2026

- Paper | Website

Evaluating Newtonian Mechanics in Video Generative Models with Real Physical Systems **ICML**

International Conference on Machine Learning | *Co-second author* 2026

- Paper | Website | Code

Reproducing NevIR: Negation in Neural Information Retrieval **SIGIR**

ACM SIGIR Conference on Research and Development in Information Retrieval | *Co-first author* 2025

- Paper | Code | Poster

Conformal Time Series Decomposition with Component-wise Exchangeability **COPA**

13th Symposium on Conformal and Probabilistic Prediction with Applications | *Co-first author* 2024

- Paper | Code | Slides

Extracurricular Activity

- World Models Reading Group: member of a reading group on world models for robotics, presenting papers on occasion Present
- AI Safety Hackathon: project on LLM interpretability, awarded 2nd place 2024
- European Robotics Forum Hackathon: implemented autonomous driving and object detection on a wheeled robot 2022